Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

Practical – III

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3.A). Create a Servlet application to upload and download a file.

HTML CODE:-

```
<!DOCTYPE html>
<html>
  <head>
    <title>File Upload</title>
  </head>
  <body>
    <h1>File Upload Application</h1>
    <form action="FileUploadServlet" method="post" enctype="multipart/form-data">
      Select file to upload: <input type="file" name="file" required>
      <br><br>
      <input type="submit" value="Upload File">
    </form>
    <h2>Download Links</h2>
    <!-- Link to download files, ensure the path is correct -->
    <a href="FileDownloadServlet?filename=drawing.pdf">Download the uploaded
      file</a><br><br><br><br>
  </body>
</html>
```

SERVLET CODE:-

FileUploadServlet.java

```
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;
import javax.servlet.ServletException;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.Part;
@WebServlet("/FileUploadServlet")
@MultipartConfig
```

```
public class FileUploadServlet extends HttpServlet {
   private static final String UPLOAD_DIR = "uploads";
   @Override
   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      String uploadPath = getServletContext().getRealPath("") + File.separator + UPLOAD_DIR;
      File uploadDir = new File(uploadPath);
      if (!uploadDir.exists()) {
         uploadDir.mkdir();
      }
      Part filePart = request.getPart("file");
      String fileName = filePart.getSubmittedFileName();
      File file = new File(uploadPath + File.separator + fileName);
      try (InputStream inputStream = filePart.getInputStream();
            OutputStream outputStream = new FileOutputStream(file)) {
         byte[] buffer = new byte[4096];
         int bytesRead;
         while ((bytesRead = inputStream.read(buffer)) != -1) {
            outputStream.write(buffer, 0, bytesRead);
         }
      }
      response.getWriter().println("File uploaded successfully.");
   }
}
```

FileDownloadServlet.java

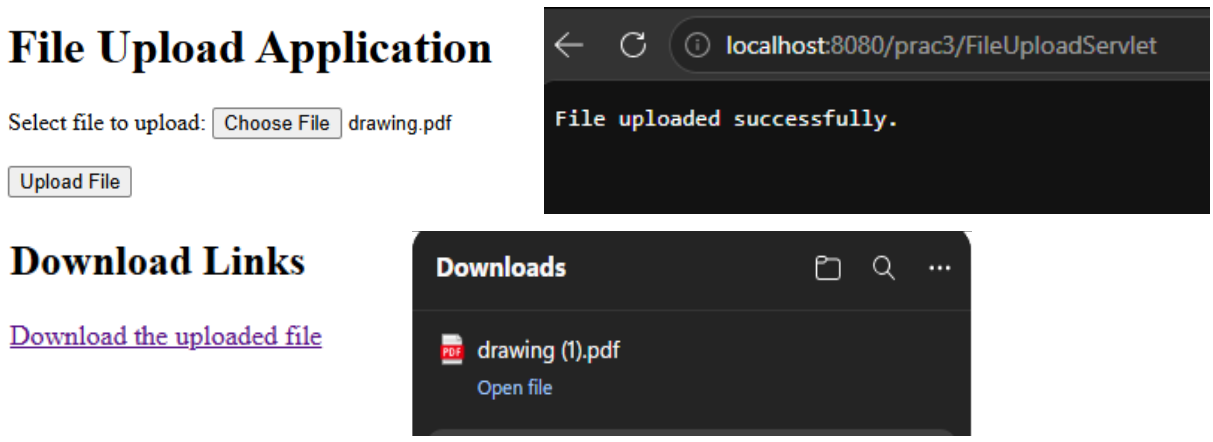
```
import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.OutputStream;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/FileDownloadServlet")
public class FileDownloadServlet extends HttpServlet {
   private static final String UPLOAD_DIR = "uploads";
   @Override
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      String fileName = request.getParameter("filename");
      if (fileName == null) {
         response.getWriter().println("File name is missing.");
         return;
      }
      String filePath = getServletContext().getRealPath("") + File.separator + UPLOAD_DIR
```

```
            + File.separator + fileName;
        File file = new File(filePath);
        if (!file.exists()) {
            response.getWriter().println("File not found.");
            return;
        }
        response.setContentType("application/octet-stream");
        response.setHeader("Content-Disposition", "attachment; filename=\"" + fileName
                + "\"");
        try (FileInputStream inputStream = new FileInputStream(file);
                OutputStream outputStream = response.getOutputStream()) {
            byte[] buffer = new byte[4096];
            int bytesRead;
            while ((bytesRead = inputStream.read(buffer)) != -1) {
                outputStream.write(buffer, 0, bytesRead);
            }
        }
    }
}
```

Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE glassfish-web-app PUBLIC "-//GlassFish.org//DTD GlassFish Application Server
3.1 Servlet 3.0//EN" "http://glassfish.org/dtds/glassfish-web-app_3_0-1.dtd">
<glassfish-web-app error-url="">
  <context-root>/FileUploadanDownload</context-root>
  <class-loader delegate="true"/>
  <jsp-config>
    <property name="keepgenerated" value="true">
      <description>Keep a copy of the generated servlet class' java code.</description>
    </property>
  </jsp-config>
</glassfish-web-app>
```

OUTPUT:-



3.B). Develop a Simple Servlet Question Answer Application using Database.

MYSQL CODE:-

```
CREATE DATABASE zeus;
USE zeus;
CREATE TABLE quiz
(
qno varchar(5) PRIMARY KEY,
question varchar(200),
op1 varchar(100),
op2 varchar(100),
op3 varchar(100),
op4 varchar(100),
ans varchar(100)
);
INSERT INTO quiz VALUES
('001','Which keyword is used to inherit a class in
Java?','implements','extends','inherit','using','extends');

INSERT INTO quiz VALUES
('002','Which method is the entry point of a Java
program?','start()','main()','run()','init()','main()');

INSERT INTO quiz VALUES
('003','Which package contains Scanner
class?','java.io','java.util','java.lang','java.sql','java.util');

INSERT INTO quiz VALUES
('004','Which is not a primitive data type?','int','float','String','char','String');

INSERT INTO quiz VALUES

('005','Which exception occurs when a number is divided by
zero?','IOException','NullPointerException','ArithmeticException','ClassNotFoundException','
ArithmeticException');
```

HTML CODE:-

```
<!DOCTYPE html>
<html>
<head>
<title>Java Quiz</title>
</head>
<body bgcolor="lightyellow">
<center>
<h1>Java Quiz Application</h1>
<h3>Click below to start the quiz</h3>
<a href="QuizServlet">START QUIZ</a>
</center>
</body>
```

```
</html>
```

SERVLET CODE:-

QuizServlet.java

```
package mypack;
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet("/QuizServlet")
public class QuizServlet extends HttpServlet {
   @Override
   protected void doGet(HttpServletRequest request,
         HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      out.println("<html>");
      out.println("<body bgcolor='lightblue'>");
      out.println("<center>");
      out.println("<h1>Java Quiz</h1>");
      out.println("</center>");
      out.println("<form action='ShowResult' method='get'>");
      try {
         Class.forName("com.mysql.jdbc.Driver");
         System.out.println("Driver Loaded Successfully");
         try (Connection con = DriverManager.getConnection(
               "jdbc:mysql://localhost:3306/zeus",
               "root",
               "root")) {
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM quiz");
            int i = 1;
            while (rs.next()) {
               out.println("<hr>");
               out.println("<h3>Q" + i + ". "
                     + rs.getString("question")
                     + "</h3>");
               out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op1") + "'>
" + rs.getString("op1") + "<br>");
```

```
                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op2") + "'>
" + rs.getString("op2") + "<br>");
                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op3") + "'>
" + rs.getString("op3") + "<br>");
                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op4") + "'>
" + rs.getString("op4") + "<br>");
                i++;
             }
          }
       } catch (Exception e) {
          out.println(e);
       }
       out.println("<br>");
       out.println("<input type='submit' value='Submit Quiz'>");
       out.println("<input type='reset' value='Reset'>");
       out.println("</form>");
       out.println("</body>");
       out.println("</html>");
   }
}
```

ShowResult.java

```
package mypack;
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet("/ShowResult")
public class ShowResult extends HttpServlet {
   @Override
   public void doGet(HttpServletRequest request,
         HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      out.println("<html>");
      out.println("<body bgcolor='white'>");
      out.println("<center>");
      out.println("<h1>Quiz Result</h1>");
      out.println("</center>");
```

```
        try {
            Class.forName("com.mysql.jdbc.Driver");
            System.out.println("Driver Loaded Successfully");
            try (Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/zeus",
                    "root",
                    "root")) {
                Statement st = con.createStatement();
                ResultSet rs = st.executeQuery("select ans from quiz");
                int score = 0;
                int i = 1;
                while (rs.next()) {
                    String correct = rs.getString("ans");
                    String user = request.getParameter("" + i);
                    if (correct.equals(user)) {
                        score++;
                        out.println("<h3>Q" + i + " : Correct</h3>");
                    } else {
                        out.println("<h3>Q" + i + " : Incorrect</h3>");
                    }
                    i++;
                }
                out.println("<hr>");
                out.println("<h2>Your Score : " + score + " / " + (i - 1) + "</h2>");
            }
        } catch (ClassNotFoundException | SQLException e) {
            out.println(e);
        }
        out.println("<br><a href='index.html'>Take Quiz Again</a>");
        out.println("</body>");
        out.println("</html>");
    }
}
```

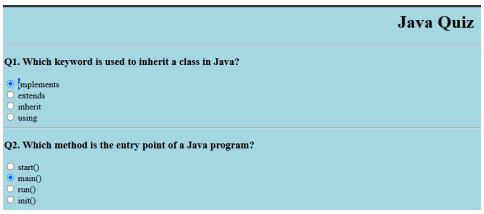
Web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
         version="3.1">
```

```
    <servlet>
        <servlet-name>QuizServlet</servlet-name>
        <servlet-class>mypack.QuizServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>QuizServlet</servlet-name>
        <url-pattern>/QuizServlet</url-pattern>
    </servlet-mapping>
    <servlet>
        <servlet-name>ShowResult</servlet-name>
        <servlet-class>mypack.ShowResult</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>ShowResult</servlet-name>
        <url-pattern>/ShowResult</url-pattern>
    </servlet-mapping>
</web-app>
```

OUTPUT:-









3.C). Create simple Servlet application to demonstrate Non-Blocking Read Operation.

HTML CODE:-

```
<!DOCTYPE html>
<html>
    <head>
        <title>Non-Blocking Practical</title>
    </head>
    <body>
        <form action="NBServ" method="post">
            <h3>Please Click Start Button to Start Non-Blocking Read!!!</h3>
```

```
        <input type="submit" value="Start NB IO Read">
    </form>
    <div>TO DO write content</div>
  </body>
</html>
```

SERVLET CODE:-

MyReadListener.java

```
import java.io.IOException;

import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.*;

@WebServlet(urlPatterns = {"/prac1c"})
public class prac1c extends HttpServlet {

  protected void doPost(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
    try (PrintWriter out = response.getWriter()) {
      String connectionURL = "jdbc:mysql://localhost:3306/db";
      Connection connection = null;
      ResultSet rs;
      response.setContentType("text/html;charset=UTF-8");
      String uid = request.getParameter("uid");
      String fname = request.getParameter("fname");
      String sname = request.getParameter("sname");
      String pwd = request.getParameter("pwd");
      String pwd1 = request.getParameter("pwd1");
      String town = request.getParameter("town");
      String country = request.getParameter("country");
      String zip = request.getParameter("zip");
      try {
        Class.forName("org.gjt.mm.mysql.Driver");
        connection = DriverManager.getConnection(connectionURL, "root", "root");
        String sql = "insert into user_register values(?,?,?,?,?,?,?,?)";
```

```
            PreparedStatement pst = connection.prepareStatement(sql);
            pst.setString(1, uid);
            pst.setString(2, fname);
            pst.setString(3, sname);
            pst.setString(4, pwd);
            pst.setString(5, pwd1);
            pst.setString(6, town);
            pst.setString(7, country);
            pst.setString(8, zip);
            int numRowsChanged = pst.executeUpdate();
            out.println("Welcome : ");
            out.println("""" + fname + """");
            pst.close();
         } catch (ClassNotFoundException e) {
            out.println("Couldnt Load database driver : " + e.getMessage());
         } catch (SQLException e) {
            out.println("SQL Exception caught : " + e.getMessage());
         } catch (Exception e) {
            out.println(e);
         } finally {
            try {
               if (connection != null) {
                  connection.close();
               }
            } catch (SQLException ignored) {
               out.println(ignored);
            }
         }
      }
   }
}
```

NBServ.java

```
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.IOException;
import java.io.InputStream;
import java.io.InputStreamReader;
import java.io.OutputStreamWriter;
import java.io.PrintWriter;
import java.net.HttpURLConnection;
```

```
import java.net.URL;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletContext;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(urlPatterns = {"/NBServ"})
public class NBServ extends HttpServlet {
   protected void processRequest(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      String filepath = "/WEB-INF/products.txt";
      ServletContext cnt = getServletContext();
      InputStream iStream = cnt.getResourceAsStream(filepath);
      try (PrintWriter out = response.getWriter()) {
         String path = "http://" + request.getServerName() + ":" + request.getServerPort()
               + request.getContextPath() + "/ReadNBServ";
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Non-Blocking I/O Servlet</title>");
         out.println("</head>");
         out.println("<body>");
         out.println("<h1>Reading Product Names from file</h1>");
         out.flush();
         URL url = new URL(path);
         HttpURLConnection mycon = (HttpURLConnection) url.openConnection();
         mycon.setChunkedStreamingMode(2);
         mycon.setDoOutput(true);
         mycon.connect();
         if (iStream != null) {
            InputStreamReader isr = new InputStreamReader(iStream);
            BufferedReader br = new BufferedReader(isr);
            String readText;
            try (BufferedWriter bw = new BufferedWriter(new
OutputStreamWriter(mycon.getOutputStream()))) {
               out.println("<div style='width=100%;height:450px;overflow:scroll;'>");
               while ((readText = br.readLine()) != null) {
```

```
                out.println("<div style='background-color:khaki;width=100%;'>");
                out.println(readText);
                out.println("</div><br/>");
                out.flush();
                bw.write(readText);
                Thread.sleep(1000);
                out.flush();
            }
            out.println("</div>");
            bw.flush();
        }
      }
      out.println("</body>");
      out.println("</html>");
    } catch (InterruptedException | IOException ex) {
      Logger.getLogger(NBServ.class.getName()).log(Level.SEVERE, null, ex);
    }
  }
  @Override
  protected void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
    processRequest(request, response);
  }
  @Override
  protected void doPost(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
    processRequest(request, response);
  }
  @Override
  public String getServletInfo() {
    return "Short description";
  }
}
```

ReadNBServ.java:

```
import EJPracts.MyReadListener;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.AsyncContext;
import javax.servlet.ServletException;
import javax.servlet.ServletInputStream;
```

```
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "ReadNBServ", urlPatterns = {"/ReadNBServ"}, asyncSupported = true)
public class ReadNBServ extends HttpServlet {
   protected void processRequest(HttpServletRequest request, HttpServletResponse response)
           throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         out.println("<!DOCTYPE html>");
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Servlet ReadNBServ</title>");
         out.println("</head>");
         out.println("<body>");
         AsyncContext cnt = request.startAsync();
         ServletInputStream iStream = request.getInputStream();
         iStream.setReadListener(new MyReadListener(iStream, cnt));
         out.println("</body>");
         out.println("</html>");
      }
   }
   @Override
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
           throws ServletException, IOException {
      processRequest(request, response);
   }
   @Override
   protected void doPost(HttpServletRequest request, HttpServletResponse response)
           throws ServletException, IOException {
      processRequest(request, response);
   }
   @Override
   public String getServletInfo() {
      return "Short description";
   }
}
```

<u>OUTPUT:-</u>



Please Click Start Button to Start Non-Blocking Read!!!

Start NB IO Read

TO DO write content

Reading Product Names from file

Laptop
Mouse
Keyboard
Monitor
Printer
Scanner
Webcam
Speaker
Pendrive
Hard Disk